Optimization Methods – Notes

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1 Lesson 01 — 24/09/2026

1.1 Introduction to optimization problems

The aim of the course is to give an overview, from the mathematical point of view, of the main optimization methods for data analysis and decision problems. The main tool is an optimization problem

$$\min\{f(x) : x \in X\}, \tag{P}$$

where $f : \mathbb{R}^n \to \mathbb{R}$ is the `objective function` and $X \subseteq \mathbb{R}^n$ is the `constraints set` (or `feasible region`). If $X = \mathbb{R}^n$ the problem is said to be `unconstrained`.

Remark. (P) can be seen as a decision problem: X is the set of all the admissible (feasible) decisions x and $f(x)$ is the value of the decision x (for example a cost or a gain).

In general X is described by constraint functions,

$$\begin{cases} \min f(x) \\ g(x) \leq 0 \\ h(x) = 0 \end{cases}$$

where $g = (g_1, \dots, g_m)$ are the inequality constraints functions and $h = (h_1, \dots, h_p)$ are the equality constraints functions. We will generally consider minimization problems, since $\max\{f(x) : x \in X\} = -\min\{-f(x) : x \in X\}$.

It is fundamental to analyze the properties of (P) under suitable assumptions on X and on the involved functions, mainly `convexity` and `differentiability`. This is why we start from some notions of convex analysis.

1.1.1 An application: Economical Power Dispatch

A power system has to satisfy a given electricity load demand P_D. The electricity is produced by n different units, each with a fuel cost function. The `Economical Power Dispatch` (EPD) consists in determining the load to be produced by each unit, in a given interval of time T, in order to minimize the total fuel cost while supplying the total demand and respecting some technical requirements.

The model is the following:

- p_i (in MW) is the unknown power generated by unit $i = 1, \dots, n$;
- $f_i(p_i) = a_i p_i^2 + b_i p_i + c_i$ is the fuel cost (Euro/T) of unit i, with a_i, b_i, c_i the fuel cost coefficients;
- $p_i^{\min}$ and $p_i^{\max}$ are the minimum and maximum output of unit i;
- P_D is the total demand and P_L the system losses (in MW).

$$\begin{cases} \min F(p) \coloneqq \sum_{i=1}^n f_i(p_i) \\ p_1 + p_2 + \dots + p_n = P_D + P_L \\ p_i^{\min} \leq p_i \leq p_i^{\max}, \quad i = 1, \dots, n \end{cases}$$

Remark. If we also want to minimize the emissions $E(p) = \sum_i e_i(p_i)$, with $e_i(p_i) = \alpha_i p_i^2 + \beta_i p_i + \gamma_i$, we get a `multiobjective` problem $\min(F(p), E(p))$ with the same constraints (Environmental/Economical Power Dispatch).

The course program covers: convex analysis, existence of optima, optimality conditions and duality, solution methods (gradient, conjugate gradient, Newton), applications to machine learning (SVM), multiobjective optimization and non-cooperative game theory.

1.2 Convex sets

Definition 1.1 (Convex set)**.**
A set $C \subseteq \mathbb{R}^n$ is *convex* if, for every $x, y \in C$ and every $\alpha \in [0,1]$,

$$\alpha x + (1-\alpha)y \in C.$$

In words, the segment joining any two points of C is contained in C.

Definition 1.2 (Affine set)**.**
A set $C \subseteq \mathbb{R}^n$ is *affine* if, for every $x, y \in C$ and every $\alpha \in \mathbb{R}$,

$$\alpha x + (1-\alpha)y \in C.$$

Remark. Every affine set is convex, since $[0,1] \subset \mathbb{R}$.

Examples of affine sets are: any single point $\{x\}$, any line, any subspace, the solution set of a linear system $C = \{x \in \mathbb{R}^n : Ax = b\}$, with A an $m \times n$ matrix and $b \in \mathbb{R}^m$.

A *subspace* is a particular affine set: $S \subseteq \mathbb{R}^n$ is a subspace if $\alpha x + \beta y \in S$ for every $x, y \in S$ and $\alpha, \beta \in \mathbb{R}$. Examples are $\{0\}$, any line through zero, the solution set of a homogeneous system $\{x : Ax = 0\}$.

Definition 1.3 (Convex combination)**.**
A *convex combination* of the points $x^1, x^2, \dots, x^k$ is a point

$$y = \sum_{i=1}^{k} \alpha_i x^i \quad \text{where} \quad \alpha_1, \dots, \alpha_k \in [0,1] \quad \text{and} \quad \sum_{i=1}^{k} \alpha_i = 1.$$

Remark. By definition, a set $C \subseteq \mathbb{R}^n$ is *convex* if it contains all the convex combinations of any two points in C.

Example 1.4.
Take $x^1 = (1,1)$, $x^2 = (3,1)$, $x^3 = (2,3)$. The point $x = (2,2)$ is a convex combination of them:

$$x = \tfrac{1}{4}x^1 + \tfrac{1}{4}x^2 + \tfrac{1}{2}x^3.$$

The next lemma says that convexity is enough to be closed under convex combinations of *any* number of points, not only two.

Lemma 1.5.
If C is convex, then for any $x^1, \dots, x^k \in C$ and $\alpha_1, \dots, \alpha_k \in [0,1]$ such that $\sum_{i=1}^k \alpha_i = 1$,

$$\sum_{i=1}^{k} \alpha_i x^i \in C.$$

Proof. By induction on k. For $k = 2$ the thesis holds by definition of convexity. Assume it holds for k and let $x^1, \dots, x^{k+1} \in C$, $\alpha_1, \dots, \alpha_{k+1} \in [0,1]$ with $\sum_{i=1}^{k+1} \alpha_i = 1$. Without loss of generality $\alpha_1 \neq 1$. Then

$$\sum_{i=1}^{k+1} \alpha_i x^i = \alpha_1 x^1 + \Big(\sum_{i=2}^{k+1} \alpha_i\Big) \sum_{i=2}^{k+1} \frac{\alpha_i}{\sum_{j=2}^{k+1} \alpha_j} x^i.$$

Since $\sum_{i=2}^{k+1} \frac{\alpha_i}{\sum_{j=2}^{k+1} \alpha_j} = 1$, by the inductive assumption $\bar{x} := \sum_{i=2}^{k+1} \frac{\alpha_i}{\sum_{j=2}^{k+1} \alpha_j} x^i \in C$. Finally, since C is convex and $\sum_{i=2}^{k+1} \alpha_i = 1 - \alpha_1$, the point $\alpha_1 x^1 + (1-\alpha_1)\bar{x}$ belongs to C. □

Proposition 1.6.
If $\{C_i\}_{i\in I}$ is any (possibly infinite) family of convex sets, then $\bigcap_{i\in I} C_i$ is convex.

This proposition makes the following definition well posed.

Definition 1.7 (Convex hull)**.**
The *convex hull* $\mathrm{conv}(C)$ of a set C is the intersection of all the convex sets containing C. In other words, it is the smallest convex set containing C.

For example, the convex hull of x^1, x^2, x^3 above is the triangle with vertices the three points.

Proposition 1.8.
$\mathrm{conv}(C) = \{$*all convex combinations of points in* $C\}$.

Proof. The set of convex combinations of points in C is a convex set containing C, hence $\mathrm{conv}(C)$ is contained in it. Conversely, $C \subseteq \mathrm{conv}(C)$ and $\mathrm{conv}(C)$ is convex, so by the previous lemma it contains every convex combination of points of C. □

Remark. C is convex if and only if $C = \mathrm{conv}(C)$.

1.2.1 Examples of convex sets

Definition 1.9 (Polyhedron)**.**
A *polyhedron* P is the intersection of a finite number of closed halfspaces in $\mathbb{R}^n$. A closed halfspace is the set of solutions of a linear inequality $a^T x \leq \beta$, with $a \in \mathbb{R}^n$, $\beta \in \mathbb{R}$. Hence

$$P = \{x \in \mathbb{R}^n : Ax \leq b\},$$

with A an $m \times n$ matrix and $b \in \mathbb{R}^m$.

A polyhedron is convex, since any closed halfspace is convex and the intersection of convex sets is convex.

Definition 1.10 (Ball)**.**
A *ball* is $B(\bar{x}, r) \coloneqq \{z \in \mathbb{R}^n : \|z - \bar{x}\| \leq r\}$, where $\|\cdot\|$ is any norm, e.g.

- $\|x\|_2 = \sqrt{\sum_i x_i^2}$ (Euclidean norm);
- $\|x\|_1 = \sum_i |x_i|$ (Manhattan distance);
- $\|x\|_\infty = \max_i |x_i|$ (Chebyshev norm);
- $\|x\|_p = \left(\sum_i |x_i|^p\right)^{1/p}$, with $1 \leq p < +\infty$;
- $\|x\|_A = \sqrt{x^T A x}$, with A symmetric and positive definite, i.e. $x^T A x > 0$ for all $x \neq 0$.

Recall that a norm on a real vector space X is a function $p : X \to \mathbb{R}$ such that $p(x + y) \leq p(x) + p(y)$, $p(\alpha x) = |\alpha| p(x)$, and $p(x) = 0 \Leftrightarrow x = 0$; these imply $p(x) \geq 0$.

Balls are convex for any norm; we prove it in the next lesson, after some examples.

2 Lesson 02 — 25/09/2026

2.1 Examples on balls

We start the lesson with some examples on balls.

Example 2.1 (Exercise 1.1)**.**
Find the unit ball $B(0,1)$ w.r.t. $\|\cdot\|_1$, $\|\cdot\|_\infty$ and $\|\cdot\|_A$, where

$$A = \begin{bmatrix} 4 & 0 \\ 0 & 1 \end{bmatrix}.$$

- For $\|x\|_1 = |x_1| + |x_2|$ the ball is $B(0,1) = \{x : |x_1| + |x_2| \leq 1\}$. Since

  $$|x_1| + |x_2| = \max\{x_1 + x_2,\ x_1 - x_2,\ -x_1 + x_2,\ -x_1 - x_2\},$$

  all four expressions must be ≤ 1, so the ball is a square rotated by 45 degrees, with vertices $(\pm 1, 0)$, $(0, \pm 1)$.
- For $\|x\|_\infty = \max\{|x_1|, |x_2|\}$ the ball is $\{x : |x_1| \leq 1,\ |x_2| \leq 1\}$, i.e. the square $[-1,1]^2$.
- For $\|x\|_A = \sqrt{x^T A x}$ we have

  $$x^T A x = \begin{bmatrix} x_1 & x_2 \end{bmatrix} \begin{bmatrix} 4 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 4x_1^2 + x_2^2 \leq 1,$$

  which is an ellipse.

All of them are convex sets.

The fact that these balls are all convex is not a coincidence.

Example 2.2 (Exercise 1.4)**.**
Prove that $B(0,1) = \{x \in \mathbb{R}^n : \|x\| \leq 1\}$ is convex for any norm.
Let $x, y \in B(0,1)$; we have to prove that $\alpha x + (1-\alpha)y \in B(0,1)$ for all $\alpha \in [0,1]$. By the triangle inequality and homogeneity,

$$\|\alpha x + (1-\alpha)y\| \leq \|\alpha x\| + \|(1-\alpha)y\| = \alpha \|x\| + (1-\alpha)\|y\| \leq \alpha + (1-\alpha) = 1.$$

2.2 Operations that preserve convexity

2.2.1 Algebraic operations

Lemma 2.3.
If C_1 and C_2 are convex, then $C_1 + C_2 \coloneqq \{x+y : x \in C_1, y \in C_2\}$ is convex. If C is convex and $\alpha \in \mathbb{R}$, then $\alpha C \coloneqq \{\alpha x : x \in C\}$ is convex. Consequently, $C_1 - C_2 \coloneqq \{x-y : x \in C_1, y \in C_2\}$ is convex.

2.2.2 Topological operations

Lemma 2.4.
If C is convex then $\mathrm{cl}(C)$ is convex. If C is convex then $\mathrm{int}(C)$ is convex, provided that $\mathrm{int}(C) \neq \emptyset$.

The interior of a convex set can be empty even when the set is "reasonable" (e.g. a segment in $\mathbb{R}^2$). For this reason it is useful to look at the interior relative to the smallest affine set containing C. We denote by $\text{aff}(C)$ the smallest affine set containing C.

Definition 2.5 (Relative interior)**.**
Let $C \subseteq \mathbb{R}^n$ be a convex set. The *relative interior* of C is

$$\text{ri}(C) = \{x \in C : \exists\, \epsilon > 0 \text{ s.t. } \text{aff}(C) \cap B(x, \epsilon) \subseteq C\}.$$

Example 2.6.
Let $C \coloneqq \{(x_1, x_2) \in \mathbb{R}^2 : 1 \leq x_1 \leq 3,\ x_2 = 0\}$. Then $\text{ri}(C) = \{(x_1, x_2) : 1 < x_1 < 3,\ x_2 = 0\}$, while $\text{int}(C) = \emptyset$. If $C = \{\bar{x}\}$, then $\text{ri}(C) = C$.

Theorem 2.7.
If C is a nonempty convex set in $\mathbb{R}^n$, then $\text{ri}(C)$ is a nonempty convex set.

2.2.3 Separation of convex sets

The relative interior is what is needed to state when two convex sets can be separated by a hyperplane.

Two sets $A, B \subseteq \mathbb{R}^n$ are *linearly separable* if there exist $a \in \mathbb{R}^n$, $a \neq 0$, and $\beta \in \mathbb{R}$ such that

$$a^T x \geq \beta \quad \forall x \in A, \qquad a^T x \leq \beta \quad \forall x \in B.$$

The separation is *proper* if the strict inequality holds for at least one $x \in A \cup B$.

Theorem 2.8 (Separation)**.**
Let A, B be nonempty convex sets in $\mathbb{R}^n$. Then A and B are properly linearly separable if and only if

$$\text{ri}(A) \cap \text{ri}(B) = \emptyset.$$

In particular, two disjoint convex sets are always properly linearly separable.

Example 2.9.
Let $A \coloneqq \{x \in \mathbb{R}^2 : 1 \leq x_1 \leq 2,\ x_2 = 0\}$ and $B \coloneqq \{x \in \mathbb{R}^2 : 2 \leq x_1 \leq 4,\ x_2 = 0\}$. Then $\text{ri}(A) \cap \text{ri}(B) = \emptyset$ and the sets are properly separable by the hyperplane $x_1 = 2$.

2.2.4 Affine transformations

Theorem 2.10.
Let $f : \mathbb{R}^n \to \mathbb{R}^m$ be affine, i.e. $f(x) = Ax + b$, with $A \in \mathbb{R}^{m \times n}$, $b \in \mathbb{R}^m$.

- *If $C \subseteq \mathbb{R}^n$ is convex, then $f(C) = \{f(x) : x \in C\}$ is convex.*
- *If $C \subseteq \mathbb{R}^m$ is convex, then $f^{-1}(C) = \{x \in \mathbb{R}^n : f(x) \in C\}$ is convex.*

Examples of affine maps are the scaling $f(x) = \alpha x$, the translation $f(x) = x + b$ and the rotation $f(x) = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} x$.

3 Lesson 03 — 01/10/2026

3.1 Cones

Definition 3.1 (Cone)**.**
A set $C \subseteq \mathbb{R}^n$ is a *cone* if, for every $x \in C$ and every $\lambda \geq 0$,

$$\lambda x \in C.$$

In other words, if C contains a point x different from 0, then it also contains the whole half-line starting at 0 and passing through x. A cone may be convex or non-convex; here are some examples.

Example 3.2.

- $\mathbb{R}^n_+$ is a convex cone.
- $\{x \in \mathbb{R}^2 : x_1 x_2 = 0\}$ is a non-convex cone.
- Given a polyhedron $P = \{x \in \mathbb{R}^n : Ax \leq b\}$, the *recession cone* of P is

  $$\text{rec}(P) = \{d \in \mathbb{R}^n : x + \alpha d \in P \text{ for any } x \in P,\ \alpha \geq 0\},$$

  i.e. the set of directions along which we never leave the polyhedron. It can be proved that $\text{rec}(P) = \{d : Ad \leq 0\}$, so it is a polyhedral cone.
- $\{x \in \mathbb{R}^3 : x_3 \geq \sqrt{x_1^2 + x_2^2}\}$ is a non-polyhedral convex cone.
- Given $\bar{x} \in \text{cl}(C) \subseteq \mathbb{R}^n$, the *tangent cone* to C at $\bar{x}$ is

  $$T_C(\bar{x}) = \Big\{d \in \mathbb{R}^n : \exists\{z_k\} \subset C,\ \exists\{t_k\} > 0,\ z_k \to \bar{x},\ t_k \to 0,\ \lim_{k\to\infty} \frac{z_k - \bar{x}}{t_k} = d\Big\}.$$

The tangent cone will be the key object in the optimality conditions for constrained problems.

Example 3.3.
Let $P_1 = \{x \in \mathbb{R}^2 : x_2 \leq x_1,\ x_2 \geq 0\}$, a polyhedral cone: $\text{rec}(P_1) = P_1$ and $T_{P_1}((0,0)) = P_1$.
Let $P_2 = \{x \in \mathbb{R}^2 : x_1 \geq 1,\ x_2 \geq 1,\ x_1 + x_2 \geq 3\}$: then $\text{rec}(P_2) = \{d : d_1 \geq 0,\ d_2 \geq 0\}$ and $T_{P_2}((1,2)) = \{d : d_1 \geq 0,\ d_1 + d_2 \geq 0\}$.

Exercises.

1.2 Let $P = \{x : Ax \leq b\}$. Prove that $\text{rec}(P) = \{d \in \mathbb{R}^n : Ad \leq 0\}$.

1.3 If C_1 and C_2 are convex, is $C_1 \cup C_2$ convex?

1.5 Write $(1,1)$ as a convex combination of $(0,0)$, $(3,0)$, $(0,2)$, $(3,2)$.

3.2 Convex functions

We now move from sets to functions: convexity of a function is expressed through convexity of a set, its epigraph.

Definition 3.4 (Convex function)**.**
Let $C \subseteq \mathbb{R}^n$ be convex. A function $f : \mathbb{R}^n \to \mathbb{R}$ is *convex* on C if, for all $x, y \in C$ and every

$\alpha \in [0,1]$,

$$f(\alpha y + (1-\alpha)x) \leq \alpha f(y) + (1-\alpha) f(x).$$

When $C = \mathbb{R}^n$ we simply say that f is convex.

Geometrically, the graph of f between x and y lies below the chord joining $(x, f(x))$ and $(y, f(y))$.

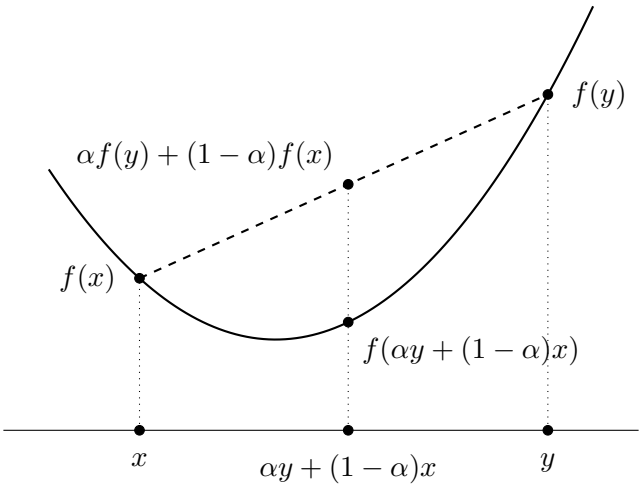


Figure 1: Example of a convex function.

Theorem 3.5.
A function $f : \mathbb{R}^n \to \mathbb{R}$ is convex if and only if the set

$$\operatorname{epi} f := \{(x, y) \in \mathbb{R}^n \times \mathbb{R} : y \geq f(x)\},$$

called the epigraph *of f, is convex.*

Definition 3.6 (Concave function)**.**
Given a convex set $C \subseteq \mathbb{R}^n$, a function $f : \mathbb{R}^n \to \mathbb{R}$ is *concave* on C if $-f$ is convex on C.

Some functions are both convex and concave, like affine functions $f(x) = c^T x + b$. Every norm $f(x) = \|x\|$ is convex.

Theorem 3.7 (Continuity of convex functions)**.**
Let $f : \mathbb{R}^n \to \mathbb{R}$ be convex on the convex set $C \subseteq \mathbb{R}^n$. Then f is continuous on $\operatorname{ri}(C)$.

3.3 Strictly and strongly convex functions

Definition 3.8 (Strictly convex function)**.**
Given a convex set $C \subseteq \mathbb{R}^n$, f is *strictly convex* on C if

$$f(\alpha y + (1-\alpha)x) < \alpha f(y) + (1-\alpha) f(x) \qquad \forall x, y \in C,\ x \neq y,\ \forall \alpha \in (0,1).$$

The only difference is that the inequality is strict: a strictly convex function cannot be affine on any segment.

Definition 3.9 (Strongly convex function)**.**
Given a convex set $C \subseteq \mathbb{R}^n$, f is *strongly convex* on C if there exists $\tau > 0$ such that, for all $x, y \in C$ and $\alpha \in [0,1]$,

$$f(\alpha y + (1-\alpha)x) \leq \alpha f(y) + (1-\alpha)f(x) - \frac{\tau}{2}\alpha(1-\alpha)\|y-x\|_2^2 .$$

Remark. Similarly, f is strictly (strongly) concave if $-f$ is strictly (strongly) convex.

Theorem 3.10.
f is strongly convex if and only if there exists $\tau > 0$ such that $f(x) - \frac{\tau}{2}\|x\|_2^2$ is convex.

Remark. Equivalently, f is strongly convex if and only if $f(x) = \psi(x) + \frac{\tau}{2}\|x\|_2^2$ for some convex function ψ and some $\tau > 0$.

Exercise 1.6. Prove that strongly convex $\Rightarrow$ strictly convex $\Rightarrow$ convex. Do the opposite implications hold?

3.3.1 First and second order characterizations

Definitions are hard to check in practice. If f is differentiable we can use derivatives. Assume from now on that $C \subseteq \mathbb{R}^n$ is open and convex.

Theorem 3.11 (First order condition)**.**
Let f be continuously differentiable on C. Then f is convex on C if and only if

$$f(y) \geq f(x) + (y-x)^T \nabla f(x) \qquad \forall x, y \in C.$$

The function $y \mapsto f(x) + (y-x)^T \nabla f(x)$ is the tangent plane to f at x, so the theorem says that f is convex if and only if it always lies above its tangent planes (the first-order approximation is a global underestimator).

The same idea characterizes strict and strong convexity.

Theorem 3.12.
Let f be continuously differentiable on C.

- *f is strictly convex on C if and only if*

$$f(y) > f(x) + (y-x)^T \nabla f(x) \qquad \forall x, y \in C,\ x \neq y.$$

- *f is strongly convex on C if and only if there exists $\tau > 0$ such that*

$$f(y) \geq f(x) + (y-x)^T \nabla f(x) + \frac{\tau}{2}\|y-x\|_2^2 \qquad \forall x, y \in C.$$

If f is twice continuously differentiable we can look at the Hessian.

Theorem 3.13 (Second order conditions)**.**
Let f be twice continuously differentiable on C.

- *f is convex on C if and only if $\nabla^2 f(x)$ is positive semidefinite for all $x \in C$, i.e.*

$$v^T \nabla^2 f(x) v \geq 0 \quad \forall v \in \mathbb{R}^n,\ \forall x \in C,$$

or, equivalently, the eigenvalues of $\nabla^2 f(x)$ are non-negative for all $x \in C$.

- *If $\nabla^2 f(x)$ is positive definite for all $x \in C$, then f is strictly convex on C.*
- *f is strongly convex on C if and only if there exists $\tau > 0$ such that $\nabla^2 f(x) - \tau I_n$ is positive semidefinite for all $x \in C$, i.e.*

$$v^T \nabla^2 f(x) v \geq \tau \|v\|_2^2 \quad \forall v \in \mathbb{R}^n, \ \forall x \in C,$$

or, equivalently, the eigenvalues of $\nabla^2 f(x)$ are greater than or equal to τ for all $x \in C$.

3.3.2 Examples and exercises

Example 3.14.
Let $f : \mathbb{R} \to \mathbb{R}$.

- $f(x) = e^{px}$, $p \neq 0$, is strictly convex on $\mathbb{R}$ but not strongly convex.
- On $C = \mathbb{R}_+ \setminus \{0\}$: $f(x) = x^p$ is strictly convex if $p > 1$ or $p < 0$, and strictly concave if $0 < p < 1$; $f(x) = \log x$ is strictly concave but not strongly concave.

Let $f : \mathbb{R}^n \to \mathbb{R}$.

- $f(x) = \|x\|$ is convex but not strictly convex.
- $f(x) = \max\{x_1, \ldots, x_n\}$ is convex but not strictly convex.

Exercises.

1.7 Prove that $f(x) = \|x\|$ is convex, whatever the norm.

1.8 Prove that if f is convex, $x^1, \ldots, x^k \in C$ and $\alpha_1, \ldots, \alpha_k \in (0,1)$ with $\sum_i \alpha_i = 1$, then $f(\sum_i \alpha_i x^i) \leq \sum_i \alpha_i f(x^i)$. (Hint: follow the proof of the lemma on convex combinations.)

1.9 Prove that $f(x_1, x_2) = \frac{1}{x_1 x_2}$ is convex on $\{x \in \mathbb{R}^2 : x_1, x_2 > 0\}$.

1.10 Analyse the convexity properties of $f(x_1, x_2, x_3) = 2x_1^2 + 2x_2^2 + x_3^2 + 3x_1x_2 + x_2x_3 - 6x_1 - 4x_2 - 3x_3$.

1.11 If f_1 and f_2 are convex, is the product $f_1 f_2$ convex?

4 Lesson 04 — 02/10/2026

4.1 Convexity of quadratic functions

A first important application of the second order conditions is the quadratic case. Let

$$f(x) = \frac{1}{2} x^T Q x + c^T x,$$

where Q is an $n \times n$ symmetric matrix and $c \in \mathbb{R}^n$. It is easy to check that $\nabla f(x) = Qx + c$, so Q is the Hessian of f. Then f is:

- convex if and only if Q is positive semidefinite;
- strongly convex if and only if Q is positive definite;
- concave if and only if Q is negative semidefinite;
- strongly concave if and only if Q is negative definite.

4.2 Operations that preserve convexity of functions

In practice, convex functions are rarely checked from the definition: we build them from simple ones using operations that preserve convexity.

Theorem 4.1.

- *If f is convex and $k > 0$, then kf is convex.*
- *If f_1 and f_2 are convex, then $f_1 + f_2$ is convex.*
- *If f is convex, then $f(Ax + b)$ is convex.*

Example 4.2.

- Log barrier for linear inequalities: $f(x) = -\sum_{i=1}^{m} \log(b_i - a_i^T x)$ on $C = \{x : b_i - a_i^T x > 0 \ \forall i\}$.
- Norm of an affine function: $f(x) = \|Ax + b\|$.

Theorem 4.3 (Pointwise maximum)**.**
If $f_1, \ldots, f_m$ are convex, then $f(x) = \max\{f_1(x), \ldots, f_m(x)\}$ is convex. More generally, if $\{f_i\}_{i \in I}$ is any family of convex functions, then $f(x) = \sup_{i \in I} f_i(x)$ is convex.

Example 4.4.
If $\psi(x, \lambda) : \mathbb{R}^n \times \mathbb{R}^m \to \mathbb{R}$ is convex in x and concave in λ, then $p(x) = \sup_\lambda \psi(x, \lambda)$ is convex and $d(\lambda) = \inf_x \psi(x, \lambda)$ is concave.

The last example will be useful when we study duality, since the dual function is an infimum of functions that are affine in the multipliers.

4.3 Optimization problem in standard form

Definition 4.5 (Optimization problem)**.**
An optimization problem in `standard form` is

$$f^* = \min\{f(x) : x \in X\}, \tag{P}$$

where

- $f : \mathbb{R}^n \to \mathbb{R}$ is the `objective function`;
- $X \subseteq \mathbb{R}^n$ is the `constraints set` or `feasible region`.

If $X \equiv \mathbb{R}^n$, then (P) is said to be `unconstrained`.

From now on, we will only consider minimization problems, since

$$\max\{f(x) : x \in X\} = -\min\{-f(x) : x \in X\}.$$

4.3.1 Mathematical programming problems

In general X is defined by the expression

$$X \coloneqq \{x \in \mathbb{R}^n : g_i(x) \leq 0,\ i \in I,\ h_j(x) = 0,\ j \in J\},$$

where $g_i : \mathbb{R}^n \to \mathbb{R}$, $i \in I \coloneqq \{1, \ldots, m\}$, and $h_j : \mathbb{R}^n \to \mathbb{R}$, $j \in J \coloneqq \{1, \ldots, p\}$.

Using the notation

$$g(x) \coloneqq (g_1(x), \ldots, g_m(x))^T, \qquad h(x) \coloneqq (h_1(x), \ldots, h_p(x))^T,$$

we can write $X = \{x \in \mathbb{R}^n : g(x) \leq 0,\ h(x) = 0\}$, with $g : \mathbb{R}^n \to \mathbb{R}^m$ and $h : \mathbb{R}^n \to \mathbb{R}^p$. The functions g and h are called the `constraint functions`.

Remark. In this case (P) is also referred to as a `mathematical programming problem`.

4.4 Global and local optima

Definition 4.6 (Optimal value)**.**
The `optimal value` of (P) is defined by

$$v(P) = \inf\{f(x) : x \in X\}.$$

- $v(P) \in \mathbb{R}$ if the problem is bounded from below;
- $v(P) = -\infty$ if the problem is unbounded from below;
- $v(P) = +\infty$ if the problem is infeasible, i.e., $X = \emptyset$.

Definition 4.7 (Global optimal solution)**.**
A `global optimal solution` of (P) is a point $x^* \in X$ such that

$$f(x^*) \leq f(x) \quad \forall x \in X.$$

The set of global minima of f on X is denoted by $X^* = \arg\min\{f(x) : x \in X\}$.

Definition 4.8 (Local optimal solution)**.**
A `local optimal solution` of (P) is a point $x^* \in X$ such that

$$f(x^*) \leq f(x) \quad \forall x \in X \cap B(x^*, r)$$

for some $r > 0$.

Example 4.9.

- $f(x) = x$, $X = \mathbb{R}$: $v(P) = -\infty$, no optimal solution.
- $f(x) = e^x$, $X = \mathbb{R}$: $v(P) = 0$, no optimal solution.
- $f(x) = x^3 - 3x$, $X = \mathbb{R}$: $v(P) = -\infty$, $x^* = 1$ is a local optimum, no global optimum.
- $f(x) = x\log(x)$, $X = \mathbb{R}_{++}$: $v(P) = -1/e$, $x^* = 1/e$ is a global optimum.
- $f(x) = 3x^4 - 8x^3 - 6x^2 + 24x + 19$, $X = \mathbb{R}$: $v(P) = 0$, $x^* = -1$ is a global optimum and $\tilde{x} = 2$ is a local optimum.

4.5 Existence of global optima

The examples above show that an optimum may fail to exist. We now give sufficient conditions for its existence.

Theorem 4.10 (Weierstrass)**.**
If the objective function f is continuous and the feasible region X is closed and bounded, then (at least) a global optimum exists.

Proof. Let $v(P) = \inf_{x\in X} f(x)$. Define a minimizing sequence $\{x^k\} \subseteq X$ such that $f(x^k) \to v(P)$. Since $\{x^k\}$ is bounded, the Bolzano–Weierstrass theorem guarantees that there exists a subsequence $\{x^{k_p}\}$ converging to some point x^*. Since X is closed, we get $x^* \in X$. Finally, $f(x^{k_p}) \to f(x^*)$ since f is continuous. Therefore $f(x^*) = v(P)$, i.e., x^* is a global optimum. □

Example 4.11.
The problem

$$\min\ x_1 + x_2, \quad x \in X \coloneqq \{(x_1, x_2) : x_1^2 + x_2^2 - 4 \le 0\}$$

admits a global optimum, since f is continuous and X is closed and bounded.

Corollary 4.12.
If the objective function f is continuous, the feasible region X is closed and there exists $k \in \mathbb{R}$ such that the k-sublevel set

$$S_k(f) = \{x \in X : f(x) \le k\} \tag{1}$$

is nonempty and bounded, then (at least) a global optimum exists.

Proof. Minimizing f on X is equivalent to minimizing f on $S_k(f)$, which is bounded and closed since f is continuous and X is closed. Then apply the Weierstrass theorem. □

Example 4.13.
The function $f(x_1, x_2) = x_1^2 + x_2^2 - 4x_1 - 2x_2$ fulfils condition (1). In fact, suitably choosing k, the set

$$x_1^2 + x_2^2 - 4x_1 - 2x_2 \le k$$

is a circle with center $C = (2, 1)$ and radius $r = \sqrt{2^2 + 1^2 + k}$.

Example 4.14.
Consider

$$\min\ e^{x_1+x_2}, \quad x \in X \coloneqq \{x_1 - x_2 \le 0,\ -2x_1 + x_2 \le 0\}.$$

f is continuous, X is closed and unbounded. But the sublevel set $S_2(f) = \{x \in X : f(x) \le 2\}$ is nonempty and bounded, thus a global optimum exists. Note that $S_2(f)$ is the solution set of the system

$$\begin{cases} x_1 + x_2 \le \log 2 \\ x_1 - x_2 \le 0 \\ -2x_1 + x_2 \le 0 \end{cases}$$

Corollary 4.15.
If the objective function f is continuous and `coercive`*, i.e.,*

$$\lim_{\substack{\|x\|\to\infty \\ x\in X}} f(x) = +\infty, \tag{2}$$

and the feasible region $X \neq \emptyset$ is closed, then (at least) a global optimum exists.

Proof. Let $\bar{x} \in X$ and $k \coloneqq f(\bar{x})$. By (2) the sublevel set $S_k(f)$ is nonempty and bounded, then apply the previous corollary to show that a global optimum exists. □

Example 4.16.
Consider

$$\min_{x \in \mathbb{R}} \; x^4 + 3x^3 - 5x^2 + x - 2.$$

Since f is coercive, there exists a global optimum.

Example 4.17.
The function $f(x_1, x_2) = x_1^2 + x_2^2 - 4x_1 - 2x_2$ fulfils condition (2), with $X = \mathbb{R}^2$. Indeed,

$$f(x_1, x_2) = x_1^2 + x_2^2 - 4x_1 - 2x_2 = (x_1 - 2)^2 - 4 + (x_2 - 1)^2 - 1.$$

If $\|x\| = \|(x_1, x_2)\| \to +\infty$, then at least one of the following conditions holds:

$$|x_1| \to +\infty \quad \text{or} \quad |x_2| \to +\infty.$$

In the first case we have

$$\lim_{\|x\| \to +\infty} f(x_1, x_2) \geq \lim_{|x_1| \to +\infty} (x_1 - 2)^2 - 5 = +\infty.$$

In the second case

$$\lim_{\|x\| \to +\infty} f(x_1, x_2) \geq \lim_{|x_2| \to +\infty} (x_2 - 1)^2 - 5 = +\infty.$$

Therefore, for any sequence $\{x^n\} = \{(x_1^n, x_2^n)\}$ such that $\|x^n\| \to +\infty$, we get

$$\lim_{n \to +\infty} f(x^n) = +\infty,$$

which guarantees that (2) is fulfilled.

5 Lesson 05 — 08/10/2026

Theorem 5.1.
Assume that f is convex on the convex set X. Then `any local optimum of (P) is a global optimum`.

Proof. Let x^* be a local optimum of (P), i.e., there is $r > 0$ s.t.

$$f(x^*) \leq f(z) \quad \forall z \in X \cap B(x^*, r).$$

By contradiction, assume that x^* is not a global optimum. Then there exists $y \in X$ such that $f(y) < f(x^*)$. Take $\alpha \in (0, 1)$ such that $\alpha x^* + (1 - \alpha) y \in B(x^*, r)$; this point belongs to X since X is convex. Then, by local optimality and convexity of f,

$$f(x^*) \leq f(\alpha x^* + (1 - \alpha) y) \leq \alpha f(x^*) + (1 - \alpha) f(y) < \alpha f(x^*) + (1 - \alpha) f(x^*) = f(x^*),$$

which is impossible. □

Proposition 5.2.
Assume that f is `strictly convex` *on the convex set X and that (P) admits a global optimum x^*. Then x^* is the* `unique` *optimal solution of (P).*

Theorem 5.3.
If $f : \mathbb{R}^n \to \mathbb{R}$ is strongly convex on $\mathbb{R}^n$ and X is closed, then there exists a global optimum.

Corollary 5.4.
If f is strongly convex (on $\mathbb{R}^n$) and X is closed and convex, then there exists a unique global optimum.

Example 5.5.
Any quadratic programming problem

$$\min \frac{1}{2} x^T Q x + c^T x, \quad x \in X,$$

where Q is a `positive definite` matrix and X is closed and convex, has a unique global optimum.

If Q is positive definite, all its eigenvalues are bounded below by a strictly positive constant $\implies$ f is strongly convex.

5.1 Optimality conditions

We now focus on optimality conditions for unconstrained problems.

Theorem 5.6 (Necessary optimality condition)**.**
Assume that X is an open set and let f be differentiable at $x^* \in X$. If x^* is a local optimum of (P), then

$$\nabla f(x^*) = 0.$$

From now on, assume that f has continuous first and second order partial derivatives for every $x \in X$.

Theorem 5.7 (Second order necessary optimality condition)**.**
Let X be an open set and let $x^* \in X$ be a local optimum for (P). Then the following conditions hold:

- $\nabla f(x^*) = 0$;
- *the Hessian matrix $\nabla^2 f(x^*)$ is positive semidefinite.*

Theorem 5.8 (Second order sufficient optimality condition)**.**
Let X be an open set, $x^* \in X$, and assume that the following conditions hold:

- $\nabla f(x^*) = 0$;
- *the Hessian matrix $\nabla^2 f(x^*)$ is positive definite.*

Then x^* is a local optimum for (P).

Theorem 5.9 (Optimality condition for convex problems)**.**
Let f be a differentiable convex function on the open convex set X. Then $x^* \in X$ is a global optimum for (P) if and only if $\nabla f(x^*) = 0$.

Proof. The necessity follows from the necessary optimality condition. It's important to prove the converse implication. If $\nabla f(x^*) = 0$, we can prove it by applying the first order condition for convex functions with $y = x^*$:

$$f(x) - f(x^*) \geq (x - x^*)^T \nabla f(x^*) = 0 \quad \forall x \in X.$$

□

Theorem 5.10.
Let f be a differentiable strictly convex function on the open convex set X. Then $x^* \in X$ is the unique global optimum for (P) if and only if $\nabla f(x^*) = 0$.

5.1.1 Existence of global optima for unconstrained quadratic programming problems

Consider the problem

$$\min f(x) := \frac{1}{2} x^T Q x + c^T x, \quad x \in \mathbb{R}^n, \tag{P}$$

where Q is an $n \times n$ symmetric matrix.

Corollary 5.11.
There exists a global optimum x^* for (P) if and only if the following conditions hold:

(i) $Qx^* + c = 0$;

(ii) Q is positive semidefinite.

Condition (i) is the same as $\nabla f(x^*) = 0$ (first order condition). Condition (ii) is the second order condition (Q is the Hessian of f).

Remark. We already observed that if Q is positive definite, then (P) admits a unique global optimum. Its expression is $x^* = -Q^{-1}c$, derived from condition (i).

If Q is positive semidefinite but not positive definite, in order to guarantee the existence of a global optimum we need to analyze the existence of solutions of the system $Qx + c = 0$ using the Rouché–Capelli theorem.

5.2 Unconstrained optimization methods

We will see some methods for optimization problems. What is an optimization methods? In an optimization descent methods we want to find a sequence of point $x^0, x^1, \ldots, x^k, \ldots$ such that $f(x^0) > f(x^1) > \ldots f(x^k) > \ldots$ and we would like that $\lim_{k\to\infty} x^k = x^*$ where x^* is such that $\nabla f(x^*) = 0$ so it's a stationary points.

A descent methods is characterize by the fact that $x^{k+1} = x^k + td^k$ where d^k is a descent direction for f at x^k and $t > 0$ is the `step`.

Definition 5.12 (descent direction)**.**
$d \in \mathbb{R}^n$ is a descent direction for f at x^k if we have that $d^T \nabla f(x^k) < 0$.

This means that the directional derivative along the direction d is strictly negative. For a suitable t sufficiently small we have that f of $f(x^{k+1}) < f(x^k)$.

How to find a descent direction? We need to solve min $d^t \nabla f(x^k)$ where the $\|d\| = 1$. In other words we want to find the steepest direction of descent. Recall that $d^t \nabla f(x^k) = \|d\| \left\| f(x^k) \right\| \cos(\theta)$ where θ is the angle between the gradient and the descent direction. To minimize that we need to minimize the cosine of θ (-1). So the solution to the min problem is $d^k = -\frac{\nabla f(x^k)}{\|\nabla f(x^k)\|}$. This is also known as the `Gradient method`.

5.2.1 Gradient method

Algorithm 1: Gradient method

Choose $x^0 \in \mathbb{R}^n$, set $k = 0$;
while $\nabla f(x^k) \neq 0$ **do**
  $d^k = -\nabla f(x^k)$; `// search direction`
  compute an optimal solution t_k of $\min_{t>0} f(x^k + td^k)$; `// step size`
  $x^{k+1} = x^k + t_k d^k$;
  $k = k + 1$;

6 Lesson 06 — 09/10/2026

6.1 Gradient method — quadratic case

If $f(x) = \frac{1}{2} x^T Q x + c^T x$, with Q a positive definite matrix, we can use the Taylor expansion at x^k:

$$f(x^k + td^k) = f(x^k) + (td^k)^T \nabla f(x^k) + \frac{1}{2}(td^k)^T Q\, td^k = \frac{1}{2}(d^k)^T Q d^k\, t^2 + (d^k)^T g^k\, t + f(x^k),$$

where $g^k = \nabla f(x^k) = Qx^k + c$. Now we want to minimize this expression with respect to t: we take the derivative with respect to t, set it equal to 0 and solve for t:

$$t_k = -\frac{(d^k)^T g^k}{(d^k)^T Q d^k}.$$

As already observed, two subsequent directions are orthogonal: $(d^k)^T d^{k+1} = 0$. This implies that the generated sequence has a zig-zag behavior, which can result in slow convergence.

Theorem 6.1 (Error bound)**.**
If $f(x) = \frac{1}{2} x^T Q x + c^T x$, with Q a positive definite matrix, and x^* is the global minimum of f, then the sequence $\{x^k\}$ satisfies the following inequality:

$$\left\| x^{k+1} - x^* \right\|_Q \leq \left(\frac{\frac{\lambda_n}{\lambda_1} - 1}{\frac{\lambda_n}{\lambda_1} + 1} \right) \left\| x^k - x^* \right\|_Q \quad \forall k \geq 0, \qquad \textit{(linear convergence)}$$

where $\|x\|_Q = \sqrt{x^T Q x}$ and $0 < \lambda_1 \leq \lambda_2 \leq \cdots \leq \lambda_n$ are the eigenvalues of Q.

Remark. If λ_n/λ_1 (the *condition number* of Q) is $\gg 1$, then the ratio

$$\frac{\frac{\lambda_n}{\lambda_1} - 1}{\frac{\lambda_n}{\lambda_1} + 1} \simeq 1$$

and the convergence may be slow.

The convergence rate is strictly related to the condition number of the matrix Q.

6.2 Inexact version of gradient methods

When f is not a quadratic function, finding the exact minimum along the direction may be computationally expensive, so in these cases we define a stopping condition on the step.

Algorithm 2: Gradient method with the Armijo inexact line search

Set $\alpha, \gamma \in (0,1)$ and $\bar{t} > 0$. Choose $x^0 \in \mathbb{R}^n$, set $k = 0$;
while $\nabla f(x^k) \neq 0$ **do**
  $d^k = -\nabla f(x^k)$, $t_k = \bar{t}$;
  while $f(x^k + t_k d^k) > f(x^k) + \alpha\, t_k (d^k)^T \nabla f(x^k)$ **do**
    $t_k = \gamma\, t_k$;
  $x^{k+1} = x^k + t_k d^k$;
  $k = k + 1$;

7 Lesson 07 — 15/10/2026

7.1 Conjugate gradient method

The `conjugate gradient method` (CG) is a descent method where the search direction involves the gradient computed at the current iteration and the direction computed at the previous iteration.

We first consider the quadratic case:

$$f(x) = \frac{1}{2} x^T Q x + c^T x,$$

where Q is positive definite. Set $g^k = \nabla f(x^k) = Qx^k + c$.

At iteration k, the search direction is defined by

$$d^k = \begin{cases} -g^0 & \text{if } k = 0, \\ -g^k + \beta_k d^{k-1} & \text{if } k \geq 1, \end{cases}$$

where β_k is such that d^k and d^{k-1} are `conjugate` with respect to Q, i.e.,

$$(d^k)^T Q d^{k-1} = 0.$$

From the previous condition we can compute β_k:

$$\beta_k = \frac{(g^k)^T Q d^{k-1}}{(d^{k-1})^T Q d^{k-1}}.$$

If we perform an exact line search, then d^k is a descent direction. The step size given by the exact line search is

$$t_k = -\frac{(g^k)^T d^k}{(d^k)^T Q d^k}.$$

Algorithm 3: Conjugate gradient method for quadratic functions

Choose $x^0 \in \mathbb{R}^n$, set $k = 0$;
while $g^k = \nabla f(x^k) \neq 0$ **do**
  if $k = 0$ **then**
    $d^k = -g^k$;
  else
    $\beta_k = \dfrac{(g^k)^T Q d^{k-1}}{(d^{k-1})^T Q d^{k-1}}$, $d^k = -g^k + \beta_k d^{k-1}$;
  $t_k = -\dfrac{(g^k)^T d^k}{(d^k)^T Q d^k}$;
  $x^{k+1} = x^k + t_k d^k$;
  $k = k + 1$;

Example 7.1.
Consider $f(x) = x_1^2 + 10x_2^2$, with starting point $x^0 = (10, 1)$.

$$Q = \begin{pmatrix} 2 & 0 \\ 0 & 20 \end{pmatrix}, \qquad \nabla f(x) = (2x_1, 20x_2).$$

- $k = 0$: $g^0 = (20, 20)$, $d^0 = -g^0 = (-20, -20)$, $t_0 = -((g^0)^T d^0)/((d^0)^T Q d^0) = 1/11$, and consequently

$$x^1 = x^0 + t_0 d^0 = (10 - 20/11, 1 - 20/11) = (90/11, -9/11).$$

- $k = 1$: $g^1 = (180/11, -180/11)$, $\beta_1 = ((g^1)^T Q d^0)/((d^0)^T Q d^0) = 81/121$, $d^1 = -g^1 + \beta_1 d^0 = (-3600/121, 360/121)$, $t_1 = -((g^1)^T d^1)/((d^1)^T Q d^1) = 11/40$, and $x^2 = x^1 + t_1 d^1 = (0, 0)$, which is the global minimum of f.

A peculiarity of this method is that for a quadratic function with positive definite Hessian it converges in at most n iterations in $\mathbb{R}^n$ (in the example, $n = 2$), as stated by the following theorem.

Theorem 7.2 (Convergence)**.**
The CG method finds the global minimum in at most n iterations. If Q has r distinct eigenvalues, then the CG method finds the global minimum in at most r iterations.

If we did not find the global minimum after k iterations, then the gradients $\{g^0, \dots, g^k\}$ are pairwise orthogonal, the directions $\{d^0, d^1, \dots, d^k\}$ are conjugate w.r.t. Q, and x^k is the minimum of f on $x^0 + \mathrm{Span}(d^0, d^1, \dots, d^{k-1})$.

Theorem 7.3 (Error bound)**.**
If $0 < \lambda_1 \leq \lambda_2 \leq \cdots \leq \lambda_n$ are the eigenvalues of Q, then the following bounds hold:

$$\left\|x^k - x^*\right\|_Q \leq 2\left(\frac{\sqrt{\lambda_n/\lambda_1} - 1}{\sqrt{\lambda_n/\lambda_1} + 1}\right)^k \left\|x^0 - x^*\right\|_Q, \qquad \forall\, k \geq 0,$$

$$\left\|x^k - x^*\right\|_Q \leq \left(\frac{\lambda_{n-k+1} - \lambda_1}{\lambda_{n-k+1} + \lambda_1}\right) \left\|x^0 - x^*\right\|_Q, \qquad \forall\, k \geq 0.$$

At iteration n, notice that the error goes to 0, since $k = n$ and $\lambda_{n-k+1} - \lambda_1 = 0$.

7.2 Newton method

7.2.1 Basic version

We want to find a stationary point of a function f, i.e., a point where $\nabla f = 0$. To study the gradient, at each iteration k we consider its linear expansion at x^k:

$$\nabla f(x) \simeq \nabla f(x^k) + \nabla^2 f(x^k)(x - x^k).$$

So, to find x^{k+1}, instead of solving $\nabla f(x) = 0$ we solve the linear system

$$\nabla f(x^k) + \nabla^2 f(x^k)(x - x^k) = 0.$$

Note that x^{k+1} is a stationary point of the quadratic approximation of f at x^k:

$$f(x) \simeq f(x^k) + (x - x^k)^T \nabla f(x^k) + \frac{1}{2}(x - x^k)^T \nabla^2 f(x^k)(x - x^k).$$

Algorithm 4: Newton method (basic version)

Choose $x^0 \in \mathbb{R}^n$, set $k = 0$;
while $\nabla f(x^k) \neq 0$ **do**
  Let d^k be the solution of the linear system $\nabla^2 f(x^k)\, d = -\nabla f(x^k)$;
  $x^{k+1} = x^k + d^k$;
  $k = k + 1$;

Theorem 7.4 (Convergence)**.**
If x^* is a local minimum of f and $\nabla^2 f(x^*)$ is positive definite, then there exists $\delta > 0$ such that for any $x^0 \in B(x^*, \delta)$ the sequence $\{x^k\}$ converges to x^* and

$$\left\|x^{k+1} - x^*\right\| \leq C \left\|x^k - x^*\right\|^2 \qquad \forall\, k > \bar{k},$$

for some $C > 0$ and $\bar{k} > 0$ (`quadratic convergence`).

Example 7.5.
The function $f(x) = 2x_1^4 + 3x_2^4 + 2x_1^2 + 4x_2^2 + x_1 x_2 - 3x_1 - 2x_2$ is strongly convex because

$$\nabla^2 f(x) = \begin{pmatrix} 24x_1^2 + 4 & 1 \\ 1 & 36x_2^2 + 8 \end{pmatrix}.$$

Starting from $x^0 = (10, 5)$, the basic Newton method gives:

k	x_1^k	x_2^k	$\left\|\nabla f(x^k)\right\|$
0	10.000000	5.000000	8189.6317378
1	6.655450	3.298838	2429.6437291
2	4.421132	2.149158	721.6330686
3	2.925965	1.361690	214.6381594
4	1.923841	0.811659	63.7752575
5	1.255001	0.428109	18.6170045
6	0.823359	0.209601	5.0058040
7	0.580141	0.171251	1.0538969
8	0.492175	0.179815	0.1022945
9	0.481639	0.180914	0.0013018
10	0.481502	0.180928	0.0000002

Drawbacks of the Newton method:

- at each iteration we need to compute both the gradient $\nabla f(x^k)$ and the Hessian matrix $\nabla^2 f(x^k)$;
- `local convergence`: if x^0 is too far from the optimum x^*, then the generated sequence may not converge to x^*.

Example 7.6.
Let $f(x) = -\frac{1}{16}x^4 + \frac{5}{8}x^2$. Then

$$f'(x) = -\frac{1}{4}x^3 + \frac{5}{4}x, \qquad f''(x) = -\frac{3}{4}x^2 + \frac{5}{4}.$$

$x^* = 0$ is a local minimum of f with $f''(x^*) = 5/4 > 0$. The sequence does not converge to x^* if it starts from $x^0 = 1$: $x^1 = -1$, $x^2 = 1$, $x^3 = -1$, ...

7.3 Newton method with line search

If f is strongly convex, then we have global convergence because d^k is a descent direction, in fact:

$$\nabla f(x^k)^T d^k = -\nabla f(x^k)^T \left[\nabla^2 f(x^k)\right]^{-1} \nabla f(x^k) < 0.$$

Algorithm 5: Newton method with (inexact) line search

Set $\alpha, \gamma \in (0,1)$ and $\bar{t} > 0$. Choose $x^0 \in \mathbb{R}^n$, set $k = 0$;
while $\nabla f(x^k) \neq 0$ **do**
  Let d^k be the solution of the linear system $\nabla^2 f(x^k)\, d = -\nabla f(x^k)$;
  $t_k = \bar{t}$;
  while $f(x^k + t_k d^k) > f(x^k) + \alpha\, t_k (d^k)^T \nabla f(x^k)$ **do**
    $t_k = \gamma\, t_k$;
  $x^{k+1} = x^k + t_k d^k$;
  $k = k + 1$;

Theorem 7.7 (Convergence)**.**
If f is strongly convex, then for any starting point $x^0 \in \mathbb{R}^n$ the sequence $\{x^k\}$ converges to the global minimum of f. Moreover, if $\alpha \in (0, 1/2)$ and $\bar{t} = 1$, then the convergence is

quadratic.

8 Lesson 08 — 16/10/2026

Lesson missed, I did it by myself using the slides.

So far we studied unconstrained problems, where a local minimum must be a stationary point. Now we move to problems with constraints: the condition $\nabla f(x^*) = 0$ is no longer necessary, and we need optimality conditions that also take the constraints into account.

8.1 First-order optimality conditions for constrained problems

Consider the constrained optimization problem

$$\min f(x), \quad x \in X := \{x \in \mathbb{R}^n : g_i(x) \leq 0,\ i = 1, \dots, m,\ h_j(x) = 0,\ j = 1, \dots, p\}, \tag{P}$$

where f, g_i and h_j are continuously differentiable for any i, j. We write $g = (g_1, \dots, g_m)$ and $h = (h_1, \dots, h_p)$, so the constraints are $g(x) \leq 0$ and $h(x) = 0$.

To describe the feasible set around a point x^* we use two cones: one built from the geometry of X and one built from the gradients of the constraints.

Definition 8.1 (Tangent cone)**.**
The `tangent cone` at $x^* \in X$ is

$$T_X(x^*) = \left\{ d \in \mathbb{R}^n : \exists \{z_k\} \subset X,\ \exists \{t_k\} > 0,\ z_k \to x^*,\ t_k \to 0,\ \lim_{k \to \infty} \frac{z_k - x^*}{t_k} = d \right\}.$$

So $T_X(x^*)$ contains the directions along which we can approach x^* while staying in X.

Definition 8.2 (Active constraints)**.**
The set of `active constraints` at $x^* \in X$ is

$$A(x^*) = \{i : g_i(x^*) = 0\},$$

i.e., the inequality constraints satisfied with equality at x^*.

Definition 8.3 (First-order feasible direction cone)**.**
The `first-order feasible direction cone` at $x^* \in X$ is

$$D(x^*) = \left\{ d \in \mathbb{R}^n : \begin{array}{ll} d^T \nabla g_i(x^*) \leq 0 & \forall\, i \in A(x^*), \\ d^T \nabla h_j(x^*) = 0 & \forall\, j = 1, \dots, p \end{array} \right\}.$$

$D(x^*)$ is the linearization of the feasible set at x^*: only the active constraints matter, because the inactive ones remain satisfied in a neighborhood of x^*.

8.2 Abadie constraint qualification (ACQ)

The tangent cone is the "true" local description of X, but it is hard to compute. The cone $D(x^*)$ is easy to compute, since it only needs gradients. We would like the two cones to coincide.

Definition 8.4 (Abadie constraint qualification)**.**
We say that the `Abadie constraint qualification` (ACQ) holds at a point $x^* \in X$ if

$$T_X(x^*) = D(x^*).$$

Checking ACQ directly from the definition is not easy either, so the following theorem gives some sufficient conditions that are simple to verify.

Theorem 8.5 (Sufficient conditions for ACQ)**.** *a)* ***Affine constraints****: if g_i and h_j are affine for all $i = 1, \dots, m$ and $j = 1, \dots, p$, then ACQ holds at any $x \in X$.*

b) ***Slater condition*** *for convex problems: if g_i are convex for all $i = 1, \dots, m$, h_j are affine for all $j = 1, \dots, p$ and there exists $\bar{x} \in X$ s.t. $g(\bar{x}) < 0$ and $h(\bar{x}) = 0$, then ACQ holds at any $x \in X$.*

c) ***Linear independence*** *of the gradients of active constraints: if $x^* \in X$ and the vectors*

$$\nabla g_i(x^*) \text{ for } i \in A(x^*), \qquad \nabla h_j(x^*) \text{ for } j = 1, \dots, p$$

are linearly independent, then ACQ holds at x^*.

8.3 Karush-Kuhn-Tucker theorem

The first step is a geometric necessary condition: at a local minimum, no direction that stays in X can be a descent direction.

Lemma 8.6.
If x^* is a local minimum of (P), then

$$d^T \nabla f(x^*) \geq 0 \qquad \forall\, d \in T_X(x^*).$$

This condition uses $T_X(x^*)$, which is hard to handle. If ACQ holds, we can replace $T_X(x^*)$ with $D(x^*)$, and this leads to an algebraic system in terms of the gradients.

Theorem 8.7 (Karush-Kuhn-Tucker)**.**
If x^* is a local minimum and ACQ holds at x^*, then there exist $\lambda^* \in \mathbb{R}^m$ and $\mu^* \in \mathbb{R}^p$ s.t. (x^*, λ^*, μ^*) satisfies the ***KKT system****:*

$$\begin{cases} \nabla f(x^*) + \sum_{i=1}^m \lambda_i^* \nabla g_i(x^*) + \sum_{j=1}^p \mu_j^* \nabla h_j(x^*) = 0, \\ \lambda_i^* g_i(x^*) = 0 \quad \forall\, i = 1, \dots, m, \\ \lambda^* \geq 0, \\ g(x^*) \leq 0, \\ h(x^*) = 0. \end{cases}$$

The vectors λ^* and μ^* are called `KKT multipliers`. The KKT system can be written in a compact form using the following function.

Definition 8.8 (Lagrangian function)**.**
The `Lagrangian function` $L : \mathbb{R}^n \times \mathbb{R}^m \times \mathbb{R}^p \to \mathbb{R}$ is

$$L(x, \lambda, \mu) := f(x) + \sum_{i=1}^m \lambda_i g_i(x) + \sum_{j=1}^p \mu_j h_j(x).$$

Then the KKT system can be formulated as:

$$\begin{cases} \nabla_x L(x, \lambda, \mu) = 0, \\ \lambda_i g_i(x) = 0, \quad i = 1, \dots, m, \\ \lambda \geq 0,\ h(x) = 0,\ g(x) \leq 0. \end{cases}$$

Remark. Since $\lambda \geq 0$ and $g(x) \leq 0$, every term $\lambda_i g_i(x)$ is ≤ 0. So the condition $\lambda_i g_i(x) = 0$ for $i = 1, \ldots, m$ is equivalent to $\lambda^T g(x) = 0$, or also $\langle \lambda, g(x) \rangle = 0$ (`complementarity condition`).

Example 8.9.
Find the minimum points of the function $f(x_1, x_2) = 2x_1 + x_2$ on the set

$$X = \{(x_1, x_2) \in \mathbb{R}^2 : x_1^2 + x_2^2 \leq 4\}.$$

Note that X is compact, so from the Weierstrass theorem f admits a global maximum and a global minimum on X.
The Lagrangian function is

$$L(x_1, x_2, \lambda) = 2x_1 + x_2 + \lambda(x_1^2 + x_2^2 - 4).$$

The KKT system is given by:

$$\begin{cases} 2 + 2\lambda x_1 = 0, \\ 1 + 2\lambda x_2 = 0, \\ \lambda(x_1^2 + x_2^2 - 4) = 0, \\ x_1^2 + x_2^2 \leq 4, \\ \lambda \geq 0. \end{cases}$$

Note that for $\lambda = 0$ the system is impossible (the first equation becomes $2 = 0$). So $\lambda > 0$, the constraint is active, and we are led to solve the system:

$$\begin{cases} x_1 = -\dfrac{1}{\lambda}, \\ x_2 = -\dfrac{1}{2\lambda}, \\ x_1^2 + x_2^2 = 4, \\ \lambda \geq 0. \end{cases}$$

Then

$$\frac{1}{\lambda^2} + \frac{1}{4\lambda^2} = 4,$$

from which

$$16\lambda^2 = 5, \qquad \lambda = \pm\frac{\sqrt{5}}{4}.$$

Since $\lambda \geq 0$, we only keep $\lambda = \frac{\sqrt{5}}{4}$:

$$x_1 = -\frac{4}{\sqrt{5}} = -\frac{4\sqrt{5}}{5}, \qquad x_2 = -\frac{2}{\sqrt{5}} = -\frac{2\sqrt{5}}{5}.$$

The global minimum exists and, as shown in the remark below, ACQ holds at every point of X, so by definition 8.7 the global minimum must solve the KKT system. Since the KKT system has only one solution, it follows that $\bar{x} = \left(-\frac{4\sqrt{5}}{5}, -\frac{2\sqrt{5}}{5}\right)$ is the global minimum point.

Remark. In the previous example ACQ is fulfilled for every $x \in X$. Indeed, there is only the constraint $g(x) = x_1^2 + x_2^2 - 4 \leq 0$, with $\nabla g(x_1, x_2) \neq (0, 0)$ for every $x \in X$ s.t. $x_1^2 + x_2^2 - 4 = 0$. So condition c) of the sufficient conditions for ACQ holds.

What about the maximum points of f on X? It is enough to set $\lambda \leq 0$ in the KKT system.

In fact, to find the maxima of f we look for the minima of $-f$. The KKT system for the problem

$$\min\left(-f(x)\right), \quad x \in X$$

is

$$\begin{cases} -2 + 2\lambda x_1 = 0, \\ -1 + 2\lambda x_2 = 0, \\ \lambda(x_1^2 + x_2^2 - 4) = 0, \\ x_1^2 + x_2^2 \leq 4, \\ \lambda \geq 0, \end{cases}$$

which is equivalent to setting $\lambda \leq 0$ in the original one. Choosing $\lambda = -\frac{\sqrt{5}}{4}$ in the original system we obtain

$$x_1 = \frac{4}{\sqrt{5}} = \frac{4\sqrt{5}}{5}, \qquad x_2 = \frac{2}{\sqrt{5}} = \frac{2\sqrt{5}}{5},$$

so that $\hat{x} = \left(\frac{4\sqrt{5}}{5}, \frac{2\sqrt{5}}{5}\right)$ is the global maximum point of f on X.

8.3.1 Limits of the KKT theorem

The KKT theorem has two limits: it needs ACQ, and it only gives necessary conditions.

Remark. The ACQ assumption is crucial in the KKT theorem.

Example 8.10.
Consider

$$\begin{cases} \min\ x_1 + x_2 \\ (x_1 - 1)^2 + (x_2 - 1)^2 - 1 \leq 0 \\ x_2 \leq 0 \end{cases}$$

The feasible set is the single point $x^* = (1, 0)$, which is therefore the global optimum.
$T_X(x^*) = \{0\}$ and $D(x^*) = \{d \in \mathbb{R}^2 : d_2 = 0\}$, hence ACQ does not hold at x^*.
$\nabla g_1(x^*) = (0, -2)$, $\nabla g_2(x^*) = (0, 1)$, $\nabla f(x^*) = (1, 1)$, hence there is no λ^* s.t. (x^*, λ^*) solves the KKT system: the first component of $\nabla_x L$ is always $1 \neq 0$.

The KKT theorem gives necessary optimality conditions, but not sufficient ones.

Example 8.11.
Consider

$$\begin{cases} \min\ x_1^2 - x_2^2 \\ x_1^2 + x_2^2 - 2 \leq 0 \end{cases}$$

$x^* = (0, 0)$, $\lambda^* = 0$ solves the KKT system, but x^* is not a local optimum.

For convex problems, instead, the KKT conditions are also sufficient.

Theorem 8.12.
If the optimization problem is convex and (x^*, λ^*, μ^*) solves the KKT system, then x^* is a global optimum.

Recall that (P) is convex if f and g are convex and h is affine.
Exercise 4.1. Prove definition 8.12.

9 Lesson 09 — 22/10/2026

9.1 Lagrangian duality

9.1.1 Lagrangian relaxation and dual function

The Lagrangian function is also the starting point of duality theory. The idea is to remove the constraints and move them into the objective function, weighted by the multipliers. Denote by $v(P)$ the optimal value of (P).

Definition 9.1 (Lagrangian relaxation)**.**
Given $\lambda \geq 0$ and $\mu \in \mathbb{R}^p$, the problem

$$\inf L(x, \lambda, \mu), \quad x \in \mathbb{R}^n$$

is called `Lagrangian relaxation` of (P), and

$$\varphi(\lambda, \mu) = \inf_{x \in \mathbb{R}^n} L(x, \lambda, \mu)$$

is the `Lagrangian dual function`.

The Lagrangian relaxation provides a lower bound to $v(P)$.

Theorem 9.2.
For any $\lambda \geq 0$ and $\mu \in \mathbb{R}^p$, we have $\varphi(\lambda, \mu) \leq v(P)$.

Proof. If $x \in X$, i.e., $g(x) \leq 0$ and $h(x) = 0$, then

$$L(x, \lambda, \mu) = f(x) + \sum_{i=1}^{m} \lambda_i g_i(x) + \sum_{j=1}^{p} \mu_j h_j(x) \leq f(x),$$

hence

$$\varphi(\lambda, \mu) = \inf_{x \in \mathbb{R}^n} L(x, \lambda, \mu) \leq \inf_{x \in X} L(x, \lambda, \mu) \leq \inf_{x \in X} f(x) = v(P). \qquad \square$$

The dual function φ has the following properties:

- it is concave, because it is the infimum of affine functions w.r.t. (λ, μ);
- it may be equal to $-\infty$ at some point;
- it may be not differentiable at some point.

9.1.2 Lagrangian dual problem

Each choice of (λ, μ) gives a lower bound, so it is natural to look for the best one.

Definition 9.3 (Lagrangian dual problem)**.**
The problem

$$\max \varphi(\lambda, \mu), \quad \lambda \geq 0, \ \mu \in \mathbb{R}^p \tag{D}$$

is called `Lagrangian dual problem` of (P), and (P) is called `primal problem`.

- The dual problem (D) consists in finding the best lower bound of $v(P)$.
- (D) is always equivalent to a convex problem, even if (P) is non-convex: indeed, it is the maximization of a concave function on a convex set.

Definition 9.2 can be equivalently stated as follows.

Theorem 9.4 (Weak duality)**.**
For any optimization problem (P), we have $v(D) \leq v(P)$.

The previous inequality is called `weak duality property`.

Example 9.5 (Linear programming)**.**
Primal problem:

$$\begin{cases} \min\ c^T x \\ Ax \geq b \end{cases} \tag{P}$$

Lagrangian function: $L(x, \lambda) = c^T x + \lambda^T (b - Ax) = \lambda^T b + (c^T - \lambda^T A)x$.
Dual function:

$$\varphi(\lambda) = \inf_{x \in \mathbb{R}^n} L(x, \lambda) = \begin{cases} -\infty & \text{if } c^T - \lambda^T A \neq 0, \\ \lambda^T b & \text{if } c^T - \lambda^T A = 0. \end{cases}$$

Dual problem:

$$\begin{cases} \max\ \varphi(\lambda) \\ \lambda \geq 0 \end{cases} \longrightarrow \begin{cases} \max\ \lambda^T b \\ \lambda^T A = c^T \\ \lambda \geq 0 \end{cases} \tag{D}$$

which is a linear programming problem.

Exercise 4.2. Find the dual of (D).

Example 9.6 (Least norm solution of linear equations)**.**
Primal problem:

$$\begin{cases} \min\ \frac{1}{2} x^T x \\ Ax = b \end{cases} \tag{P}$$

Lagrangian function: $L(x, \mu) = \frac{1}{2} x^T x + \mu^T (b - Ax)$.
Dual function: $\varphi(\mu) = \inf_{x \in \mathbb{R}^n} L(x, \mu)$.
$L(x, \mu)$ is quadratic and strongly convex with respect to x, therefore

$$\varphi(\mu) = \inf_{x \in \mathbb{R}^n} L(x, \mu) = \min_{x \in \mathbb{R}^n} L(x, \mu),$$

thus the global optimum is the stationary point:

$$\nabla_x L = x - A^T \mu = 0 \iff x = A^T \mu,$$

hence $\varphi(\mu) = -\frac{1}{2} \mu^T A A^T \mu + b^T \mu$.
Dual problem:

$$\begin{cases} \max\ -\frac{1}{2} \mu^T A A^T \mu + b^T \mu \\ \mu \in \mathbb{R}^p \end{cases} \tag{D}$$

which is an unconstrained convex quadratic programming problem.

Exercise 4.3. Find the dual problem of a general convex quadratic programming problem

$$\begin{cases} \min\ \frac{1}{2} x^T Q x + c^T x \\ Ax \leq b \end{cases} \tag{P}$$

where Q is a symmetric positive definite matrix.

9.1.3 Strong duality

Weak duality says that $v(D) \leq v(P)$. The interesting case is when the two values coincide.

Definition 9.7 (Strong duality)**.**
We say that `strong duality` holds for (P) if $v(D) = v(P)$ and (D) admits an optimal solution.

Strong duality does not hold in general.

Example 9.8.
Consider the following (non-convex) problem:

$$\begin{cases} \min\ -x^2 \\ x - 1 \leq 0 \\ -x \leq 0 \end{cases} \tag{P}$$

It is easy to check that $v(P) = -1$. The Lagrangian function is $L(x, \lambda) = -x^2 + \lambda_1(x-1) - \lambda_2 x$, hence

$$\varphi(\lambda) = \inf_{x \in \mathbb{R}} L(x, \lambda) = -\infty \qquad \forall\, (\lambda_1, \lambda_2) \in \mathbb{R}^2,$$

so $v(D) = -\infty$.

The next theorem provides sufficient conditions which guarantee strong duality for (P).

Theorem 9.9.
Suppose f, g, h are continuously differentiable, the primal problem

$$\begin{cases} \min\ f(x) \\ g(x) \leq 0 \\ h(x) = 0 \end{cases} \tag{P}$$

is convex, there exists a global optimum x^* and ACQ holds at x^*. Then:

- *strong duality holds ($v(D) = v(P)$ and (D) admits an optimal solution);*
- *(λ^*, μ^*) is optimal for (D) if and only if (λ^*, μ^*) is a KKT multipliers vector associated with x^*.*

Proof. $L(x, \lambda, \mu)$ is convex with respect to x since (P) is convex. Since ACQ holds at x^*, by definition 8.7 there exists a KKT vector of multipliers associated with x^*. Let (λ^*, μ^*) be any such vector. Then

$$\nabla_x L(x^*, \lambda^*, \mu^*) = 0, \qquad \lambda^* \geq 0, \qquad (\lambda^*)^T g(x^*) = 0.$$

Thus, using that L is convex in x (so a stationary point is a global minimum) and weak duality,

$$\begin{aligned} v(D) \geq \varphi(\lambda^*, \mu^*) &= \inf_x L(x, \lambda^*, \mu^*) = L(x^*, \lambda^*, \mu^*) \\ &= f(x^*) + (\lambda^*)^T g(x^*) + (\mu^*)^T h(x^*) = f(x^*) = v(P) \geq v(D). \end{aligned}$$

Therefore $v(P) = v(D)$ and (λ^*, μ^*) is optimal for (D).

Vice versa, if (λ^*, μ^*) is any optimal solution for (D), then

$$\begin{aligned} f(x^*) = v(P) = v(D) = \varphi(\lambda^*, \mu^*) &= \inf_x L(x, \lambda^*, \mu^*) \leq L(x^*, \lambda^*, \mu^*) \\ &= f(x^*) + (\lambda^*)^T g(x^*) + (\mu^*)^T h(x^*) \\ &= f(x^*) + (\lambda^*)^T g(x^*) \leq f(x^*), \end{aligned}$$

thus $(\lambda^*)^T g(x^*) = 0$ and $\inf_x L(x, \lambda^*, \mu^*) = L(x^*, \lambda^*, \mu^*)$, hence $\nabla_x L(x^*, \lambda^*, \mu^*) = 0$, i.e., (λ^*, μ^*) is a KKT multipliers vector associated with x^*. □

Convexity is sufficient but not necessary: strong duality may hold also for some non-convex problems.

Example 9.10.
Consider the (non-convex) problem

$$\begin{cases} \min\ -x_1^2 - x_2^2 \\ x_1^2 + x_2^2 - 1 \leq 0 \end{cases}$$

We have $v(P) = -1$. The Lagrangian function is

$$L(x, \lambda) = -x_1^2 - x_2^2 + \lambda(x_1^2 + x_2^2 - 1) = (\lambda - 1)x_1^2 + (\lambda - 1)x_2^2 - \lambda.$$

The dual function is

$$\varphi(\lambda) = \inf_{x \in \mathbb{R}^2} L(x, \lambda) = \begin{cases} -\infty & \text{if } \lambda < 1, \\ -\lambda & \text{if } \lambda \geq 1. \end{cases}$$

The dual problem is

$$\begin{cases} \max\ -\lambda \\ \lambda \geq 1 \end{cases}$$

hence its optimal solution is $\lambda^* = 1$ and $v(D) = -1 = v(P)$.

Finally, strong duality can be characterized through the saddle points of the Lagrangian function.

Theorem 9.11 (Characterization of strong duality)**.**
(x^*, λ^*, μ^*) *is a* ***saddle point*** *of L, i.e.,*

$$L(x^*, \lambda, \mu) \leq L(x^*, \lambda^*, \mu^*) \leq L(x, \lambda^*, \mu^*) \qquad \forall\, x \in \mathbb{R}^n,\ \lambda \in \mathbb{R}^m_+,\ \mu \in \mathbb{R}^p,$$

if and only if x^* is optimal for (P), (λ^*, μ^*) is optimal for (D) and $v(P) = v(D)$.

Proof. If (x^*, λ^*, μ^*) is a saddle point of L, then we can prove that $x^* \in X$ and $\langle \lambda^*, g(x^*) \rangle = 0$, which implies $\varphi(\lambda^*, \mu^*) = f(x^*)$.

Vice versa, we have that

$$f(x^*) = \varphi(\lambda^*, \mu^*) = \inf_{x \in \mathbb{R}^n} L(x, \lambda^*, \mu^*) \leq L(x^*, \lambda^*, \mu^*) = f(x^*) + \langle \lambda^*, g(x^*) \rangle,$$

hence $\langle \lambda^*, g(x^*) \rangle = 0$ and

$$L(x^*, \lambda^*, \mu^*) = f(x^*) = \varphi(\lambda^*, \mu^*) \leq L(x, \lambda^*, \mu^*) \qquad \forall\, x \in \mathbb{R}^n.$$

Moreover

$$L(x^*, \lambda, \mu) = f(x^*) + \langle \lambda, g(x^*) \rangle \leq f(x^*) = L(x^*, \lambda^*, \mu^*)$$

for all $\lambda \in \mathbb{R}^m_+$ and $\mu \in \mathbb{R}^p$. □

10 Lesson 10 — 23/10/2026

10.1 Support Vector Machine SVM

`SVM` provides a supervised classification method for vectors of data: we have a set of labeled objects and we want to assess the class of a new object.

10.1.1 Linear SVM

Linear SVM can be applied to binary classification, where we have only two classes, represented by two separated sets $A, B \subset \mathbb{R}^n$. Assume that A and B are *strictly linearly separable*, i.e., there is a hyperplane $H = \{x \in \mathbb{R}^n : w^\top x + b = 0\}$ such that

$$\begin{aligned} w^\top x^i + b > 0 \quad & \forall x^i \in A, \\ w^\top x^j + b < 0 \quad & \forall x^j \in B. \end{aligned} \tag{1}$$

If we have a new point we use the decision function $f(x) = \operatorname{sign}(w^\top x + b)$, i.e., $f(x) = 1$ if $w^\top x + b > 0$ and $f(x) = -1$ if $w^\top x + b < 0$.

A necessary and sufficient condition for eq. (1) to hold is that

$$\operatorname{conv}(A) \cap \operatorname{conv}(B) = \emptyset.$$

The problem is that there are infinitely many hyperplanes separating the two sets, and we need to decide the optimal one.

Definition 10.1 (Margin of separation)**.**
If H is a separating hyperplane, the `margin of separation` of H is the minimum distance between H and $A \cup B$, i.e.,

$$\rho(H) = \min_{x \in A \cup B} \frac{|w^\top x + b|}{\|w\|}.$$

We then look for the hyperplane with the maximum margin of separation.

Theorem 10.2.
Finding the separating hyperplane with the maximum margin of separation is equivalent to solving the convex quadratic programming problem

$$\begin{cases} \min\limits_{w,b} \ \dfrac{1}{2}\|w\|^2 \\ w^\top x^i + b \geq 1 \quad \forall x^i \in A, \\ w^\top x^j + b \leq -1 \quad \forall x^j \in B. \end{cases} \tag{2}$$

Proof. The distance between the hyperplanes $w^\top x + b = 1$ and $w^\top x + b = -1$ is $\frac{2}{\|w\|}$. In fact, consider a point $\hat{x}$ such that $w^\top \hat{x} + b = 1$. Its distance from the hyperplane $w^\top x + b + 1 = 0$ is

$$\frac{|w^\top \hat{x} + b + 1|}{\|w\|} = \frac{2}{\|w\|}.$$

Therefore, by minimizing $\|w\|$, we get two hyperplanes of maximum distance. □

Moreover, we will see that problem eq. (2) has a unique solution (w^*, b^*).

10.1.2 Equivalent formulation and dual problem

Let $\ell = |A \cup B|$. For any point $x^i \in A \cup B$, define the label

$$y^i = \begin{cases} 1 & \text{if } x^i \in A, \\ -1 & \text{if } x^i \in B, \end{cases} \qquad i = 1, \dots, \ell.$$

Then problem eq. (2) is equivalent to

$$\begin{cases} \min\limits_{w,b} \ \dfrac{1}{2}\|w\|^2 \\ 1 - y^i(w^\top x^i + b) \leq 0 \quad \forall i = 1, \dots, \ell. \end{cases} \tag{3}$$

Since problem eq. (3) is convex, it is useful to consider its Lagrangian dual.

The Lagrangian function is

$$\begin{aligned} L(w,b,\lambda) &= \frac{1}{2}\|w\|^2 + \sum_{i=1}^{\ell} \lambda_i (1 - y^i(w^\top x^i + b)) \\ &= \frac{1}{2}\|w\|^2 - \sum_{i=1}^{\ell} \lambda_i y^i w^\top x^i - b\sum_{i=1}^{\ell} \lambda_i y^i + \sum_{i=1}^{\ell} \lambda_i. \end{aligned}$$

- If $\sum_{i=1}^{\ell} \lambda_i y^i \neq 0$, then $\inf_{w,b} L(w,b,\lambda) = -\infty$.
- If $\sum_{i=1}^{\ell} \lambda_i y^i = 0$, then L does not depend on b and it is strongly convex w.r.t. w, so the optimal solution of $\min_w L(w,b,\lambda)$ is the (unique) stationary point

$$\nabla_w L(w,b,\lambda) = w - \sum_{i=1}^{\ell} \lambda_i y^i x^i = 0. \tag{4}$$

Note that, if eq. (4) holds and $\sum_{i=1}^{\ell} \lambda_i y^i = 0$, then

$$L(w,b,\lambda) = \frac{1}{2}w^\top w - w^\top w + \sum_{i=1}^{\ell} \lambda_i = -\frac{1}{2}w^\top w + \sum_{i=1}^{\ell} \lambda_i.$$

Therefore, the dual function is

$$\varphi(\lambda) = \begin{cases} -\infty & \text{if } \sum_{i=1}^{\ell} \lambda_i y^i \neq 0, \\ -\frac{1}{2}\sum_{i=1}^{\ell}\sum_{j=1}^{\ell} y^i y^j (x^i)^\top x^j \lambda_i \lambda_j + \sum_{i=1}^{\ell} \lambda_i & \text{if } \sum_{i=1}^{\ell} \lambda_i y^i = 0. \end{cases}$$

The dual of problem eq. (3) is

$$\begin{cases} \max\limits_{\lambda} \ -\frac{1}{2}\sum\limits_{i=1}^{\ell}\sum\limits_{j=1}^{\ell} y^i y^j (x^i)^\top x^j \lambda_i \lambda_j + \sum\limits_{i=1}^{\ell} \lambda_i \\ \sum\limits_{i=1}^{\ell} \lambda_i y^i = 0, \\ \lambda \geq 0, \end{cases} \tag{5}$$

or, in matrix form,

$$\begin{cases} \max\limits_{\lambda} \ -\frac{1}{2}\lambda^\top X^\top X \lambda + e_\ell^\top \lambda \\ \sum\limits_{i=1}^{\ell} \lambda_i y^i = 0, \\ \lambda \geq 0, \end{cases}$$

where $X = (y^1 x^1, y^2 x^2, \ldots, y^\ell x^\ell)$ is an $n \times \ell$ matrix and $e_\ell^\top = (1, \ldots, 1) \in \mathbb{R}^\ell$.

Remark.

- Since $X^\top X$ is always positive semidefinite, the dual problem is a convex quadratic programming problem.
- A KKT multiplier λ^* associated to the primal optimum (w^*, b^*) is a dual optimum.
- If $\lambda_i^* > 0$, then x^i is called a `support vector`.

- If λ^* is a dual optimum, then by eq. (4) we have

$$w^* = \sum_{i=1}^{\ell} \lambda_i^* y^i x^i.$$

- b^* is obtained using the complementarity conditions

$$\lambda_i^* (1 - y^i((w^*)^\top x^i + b^*)) = 0.$$

  In fact, if i is such that $\lambda_i^* > 0$, then $b^* = \frac{1}{y^i} - (w^*)^\top x^i$.

- This allows us to find the separating hyperplane $(w^*)^\top x + b^* = 0$ and the decision function $f(x) = \text{sign}((w^*)^\top x + b^*)$.

11 Lesson 11 — 29/10/2026

11.1 Linear SVM with soft margin

What if sets A and B are not linearly separable?
The linear system

$$1 - y^i(w^\top x^i + b) \leq 0 \qquad i = 1, \dots, \ell$$

has no solutions. We add some `slack variables` $\xi_i \geq 0$, as few as possible, and we consider the relaxed system

$$\begin{aligned} 1 - y^i(w^\top x^i + b) &\leq \xi_i \qquad i = 1, \dots, \ell, \\ \xi_i &\geq 0 \qquad i = 1, \dots, \ell. \end{aligned}$$

If x^i is misclassified, then $\xi_i > 1$, thus $\sum_{i=1}^{\ell} \xi_i$ is an upper bound of the number of misclassified points. In fact:

- $x^i \in A$ with $w^\top x^i + b < 0$ implies $1 < 1 - (w^\top x^i + b) = 1 - y^i(w^\top x^i + b) \leq \xi_i$;
- $x^i \in B$ with $w^\top x^i + b > 0$ implies $1 < 1 + (w^\top x^i + b) = 1 - y^i(w^\top x^i + b) \leq \xi_i$.

We redefine the objective function by adding the term $C \sum_{i=1}^{\ell} \xi_i$, where $C > 0$ is a parameter, that represents the error we make:

$$\begin{cases} \min\limits_{w,b,\xi} \dfrac{1}{2} \|w\|^2 + C \displaystyle\sum_{i=1}^{\ell} \xi_i \\ 1 - y^i(w^\top x^i + b) \leq \xi_i \quad \forall i = 1, \dots, \ell, \\ \xi_i \geq 0 \quad \forall i = 1, \dots, \ell. \end{cases} \tag{6}$$

Example 11.1 (Exercise 5.2)**.**
The dual problem of eq. (6) is

$$\begin{cases} \max\limits_{\lambda} \; -\dfrac{1}{2} \displaystyle\sum_{i=1}^{\ell} \sum_{j=1}^{\ell} y^i y^j (x^i)^\top x^j \lambda_i \lambda_j + \sum_{i=1}^{\ell} \lambda_i \\ \displaystyle\sum_{i=1}^{\ell} \lambda_i y^i = 0, \\ 0 \leq \lambda_i \leq C \quad i = 1, \dots, \ell. \end{cases} \tag{7}$$

Proof. The Lagrangian function is

$$\begin{aligned} L(w,b,\xi,\lambda,\mu) &= \frac{1}{2}\|w\|^2 + C\sum_{i=1}^{\ell}\xi_i + \sum_{i=1}^{\ell}\lambda_i(1-y^i(w^\top x^i+b)-\xi_i) - \sum_{i=1}^{\ell}\mu_i\xi_i \\ &= \frac{1}{2}\|w\|^2 - \sum_{i=1}^{\ell}\lambda_i y^i w^\top x^i - b\sum_{i=1}^{\ell}\lambda_i y^i + \sum_{i=1}^{\ell}\lambda_i + \sum_{i=1}^{\ell}\xi_i(C-\lambda_i-\mu_i). \end{aligned}$$

For fixed $(\lambda,\mu)\in\mathbb{R}^\ell_+\times\mathbb{R}^\ell_+$, L is convex w.r.t. (w,b,ξ), so a global minimum is a stationary point, i.e., a solution of

$$\begin{cases} \nabla_w L = w - \sum_{i=1}^{\ell}\lambda_i y^i x^i = 0, \\ \nabla_b L = -\sum_{i=1}^{\ell}\lambda_i y^i = 0, \\ \nabla_{\xi_i} L = C-\lambda_i-\mu_i = 0 \quad i=1,\dots,\ell. \end{cases} \tag{8}$$

If eq. (8) does not hold, then $\inf L = -\infty$. Otherwise the last sum in L vanishes and, eliminating w with the same arguments used for the dual eq. (5), we get

$$\begin{cases} \max\limits_{\lambda,\mu} \; -\frac{1}{2}\sum_{i=1}^{\ell}\sum_{j=1}^{\ell} y^i y^j (x^i)^\top x^j \lambda_i\lambda_j + \sum_{i=1}^{\ell}\lambda_i \\ \sum_{i=1}^{\ell}\lambda_i y^i = 0, \\ C-\lambda_i-\mu_i = 0 \quad i=1,\dots,\ell, \\ \lambda\geq 0,\ \mu\geq 0. \end{cases}$$

Eliminating μ, since $\mu_i = C-\lambda_i\geq 0$, we obtain $0\leq\lambda_i\leq C$, i.e., eq. (7). □

If λ^* is optimal for eq. (7), then

$$w^* = \sum_{i=1}^{\ell}\lambda_i^* y^i x^i.$$

We find b^* by choosing i such that $0<\lambda_i^*<C$ and using the complementarity conditions

$$\begin{cases} \lambda_i^*(1-y^i((w^*)^\top x^i+b^*)-\xi_i^*) = 0, \\ \mu_i^*\xi_i^* = (C-\lambda_i^*)\xi_i^* = 0. \end{cases} \tag{9}$$

Thus $b^* = \frac{1}{y^i} - (w^*)^\top x^i$.

Remark. From eq. (9) we have

- $0\leq\lambda_i^*<C$ implies $\xi_i^*=0$;
- $0<\lambda_i^*\leq C$ implies $1-y^i((w^*)^\top x^i+b^*)-\xi_i^*=0$;
- $\xi_i^*>0$ implies $\lambda_i^*=C$.

These conditions also allow us to find the errors ξ_i^*, $i=1,\dots,\ell$.

11.2 Nonlinear SVM

If two sets A and B are not linearly separable, we ask ourselves: are they linearly separable in another space? To answer this we use a map $\varphi\colon \mathbb{R}^n \to H$, where H is a higher dimensional (maybe infinite) space called `feature space`. We try to linearly separate the images $\varphi(x^i)$, $i = 1, \dots, \ell$, in the feature space.

The primal problem is

$$\begin{cases} \min\limits_{w,b,\xi} \ \dfrac{1}{2}\|w\|^2 + C\sum\limits_{i=1}^{\ell}\xi_i \\ 1 - y^i(w^\top\varphi(x^i) + b) \le \xi_i \quad \forall i = 1, \dots, \ell, \\ \xi_i \ge 0 \quad \forall i = 1, \dots, \ell, \end{cases} \tag{10}$$

where w is a vector in a high dimensional space (maybe infinitely many variables).

The dual problem is

$$\begin{cases} \max\limits_{\lambda} \ -\dfrac{1}{2}\sum\limits_{i=1}^{\ell}\sum\limits_{j=1}^{\ell} y^i y^j \varphi(x^i)^\top\varphi(x^j)\lambda_i\lambda_j + \sum\limits_{i=1}^{\ell}\lambda_i \\ \sum\limits_{i=1}^{\ell}\lambda_i y^i = 0, \\ 0 \le \lambda_i \le C \quad \forall i = 1, \dots, \ell, \end{cases} \tag{11}$$

and the number of variables is equal to the number of training data.

Let λ^* be a solution of the dual problem. Then

$$w^* = \sum_{i=1}^{\ell}\lambda_i^* y^i \varphi(x^i).$$

By any λ_i^* such that $0 < \lambda_i^* < C$ we can find b^* by the complementarity relations, that now become

$$\begin{cases} \lambda_i^*(1 - y^i((w^*)^\top\varphi(x^i) + b^*) - \xi_i^*) = 0, \\ \mu_i^*\xi_i^* = (C - \lambda_i^*)\xi_i^* = 0. \end{cases} \tag{12}$$

Then

$$1 - y^i((w^*)^\top\varphi(x^i) + b^*) = 0 \quad \longrightarrow \quad b^* = \frac{1}{y^i} - \sum_{j=1}^{\ell}\lambda_j^* y^j \varphi(x^j)^\top\varphi(x^i).$$

The decision function is given by

$$f(x) = \operatorname{sign}\big((w^*)^\top\varphi(x) + b^*\big) = \operatorname{sign}\Big(\sum_{i=1}^{\ell}\lambda_i^* y^i \varphi(x^i)^\top\varphi(x) + b^*\Big).$$

Note that $f(x)$ depends on

- λ^*, that can be found knowing $\varphi(x^i)^\top\varphi(x^j)$;
- $\varphi(x^i)^\top\varphi(x)$;
- b^*, that can be found knowing $\varphi(x^i)^\top\varphi(x^j)$.

11.2.1 Kernel function

Definition 11.2 (Kernel function)**.**
A function $k\colon \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}$ is called `kernel` if there exists a map $\varphi\colon \mathbb{R}^n \to H$ such that

$$k(x, y) = \langle \varphi(x), \varphi(y) \rangle,$$

where $\langle \cdot, \cdot \rangle$ is a scalar product in the feature space H.

Examples of kernels:

- $k(x, y) = x^\top y$;
- $k(x, y) = (x^\top y + 1)^p$, with $p \geq 1$ *(polynomial)*;
- $k(x, y) = e^{-\gamma \|x-y\|^2}$ *(Gaussian)*;
- $k(x, y) = \tanh(\beta x^\top y + \gamma)$, with suitable β and γ.

Theorem 11.3.
If $k\colon \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}$ is a kernel and $x^1, \ldots, x^\ell \in \mathbb{R}^n$, then the matrix K defined by

$$K_{ij} = k(x^i, x^j)$$

is positive semidefinite.

The dual problem depends on the kernel k:

$$\begin{cases} \max\limits_{\lambda} \; -\dfrac{1}{2} \displaystyle\sum_{i=1}^{\ell} \sum_{j=1}^{\ell} y^i y^j k(x^i, x^j) \lambda_i \lambda_j + \sum_{i=1}^{\ell} \lambda_i \\ \displaystyle\sum_{i=1}^{\ell} \lambda_i y^i = 0, \\ 0 \leq \lambda_i \leq C \quad i = 1, \ldots, \ell. \end{cases}$$

The method is:

- choose a kernel k;
- find an optimal solution λ^* of the dual;
- choose i such that $0 < \lambda_i^* < C$ and find

$$b^* = \frac{1}{y^i} - \sum_{j=1}^{\ell} \lambda_j^* y^j k(x^i, x^j);$$

- the decision function is

$$f(x) = \operatorname{sign}\Big(\sum_{i=1}^{\ell} \lambda_i^* y^i k(x^i, x) + b^* \Big).$$

The separating surface $f(x) = 0$ is linear in the feature space and nonlinear in the input space.